

Find these terms of the sequence $\{a_n\}$, where $a_n = 2(-3)^n + 5^n$.

a) $a_0 =$

b) $a_1 =$

c) $a_4 =$

d) $a_5 =$

What is the term a_8 of the sequence $\{a_n\}$ if a_n equals

a) 2^{n-1} ? _____

b) 7 ? _____

c) $1 + (-1)^n$? _____

d) $-(-2)^n$? _____

List the first 10 terms of each of these sequences.

a) the sequence obtained by starting with 10 and obtaining each term by subtracting 3 from the previous term

b) the sequence whose n^{th} term is the sum of the first n positive integers

c) the sequence whose n^{th} term is $3^n - 2^n$

d) the sequence whose n^{th} term is $\lfloor \sqrt{n} \rfloor$

e) the sequence whose first two terms are 1 and 5 and each succeeding term is the sum of the two previous terms

f) the sequence whose n^{th} term is the largest integer whose binary expansion has n bits (Write your answer in decimal notation.)

g) the sequence whose terms are constructed sequentially as follows: start with 1, then add 1, then multiply by 1, then add 2, then multiply by 2, and so on

h) the sequence whose n th term is the largest integer k such that $k! \leq n$

Find the first six terms of the sequence defined by each of these recurrence relations and initial conditions.

a) $a_n = -2a_{n-1}, a_0 = -1$

b) $a_n = a_{n-1} - a_{n-2}, a_0 = 2, a_1 = -1$

c) $a_n = 3(a_{n-1})^2, a_0 = 1$

d) $a_n = na_{n-1} + (a_{n-2})^2, a_0 = -1, a_1 = 0$

e) $a_n = a_{n-1} - a_{n-2} + a_{n-3}, a_0 = 1, a_1 = 1, a_2 = 2$

A person deposits \$1000 in an account that yields 9% interest compounded annually.

a) Set up a recurrence relation for the amount in the account at the end of n years.

b) Find an explicit formula for the amount in the account at the end of n years.

c) How much money will the account contain after 10 years?

Find the solution to each of these recurrence relations and initial conditions. Use an iterative approach such as that used in Example 10.

a) $a_n = 3a_{n-1}, a_0 = 2$

b) $a_n = a_{n-1} + 2, a_0 = 3$

c) $a_n = a_{n-1} + n, a_0 = 1$

d) $a_n = a_{n-1} + 2n + 3, a_0 = 4$

e) $a_n = 2a_{n-1} - 1, a_0 = 1$

f) $a_n = 3a_{n-1} + 1, a_0 = 1$

g) $a_n = na_{n-1}, a_0 = 5$

h) $a_n = 2na_{n-1}, a_0 = 1$

For each of these lists of integers, provide a simple formula or rule that generates the terms of an integer sequence that begins with the given list. Assuming that your formula or rule is correct, determine the next three terms of the sequence.

a) 1, 0, 1, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, . . .

b) 1, 2, 2, 3, 4, 4, 5, 6, 6, 7, 8, 8, . . .

c) 1, 0, 2, 0, 4, 0, 8, 0, 16, 0, . . .

d) 3, 6, 12, 24, 48, 96, 192, . . .

e) 15, 8, 1, -6, -13, -20, -27, . . .

f) 3, 5, 8, 12, 17, 23, 30, 38, 47, . . .

g) 2, 16, 54, 128, 250, 432, 686, . . .

h) 2, 3, 7, 25, 121, 721, 5041, 40321, . . .

What are the values of the following sums?

$$\sum_{k=1}^5 (k + 1)$$

$$\sum_{j=0}^8 (2^{j+1} - 2^j)$$

$$\sum_{j \in S} j \quad \text{where } S = \{1, 3, 4, 7, 9, 11\}$$

$$\sum_{j=0}^8 3 \cdot 2^j$$

$$\sum_{j=0}^8 (2 \cdot 3^j + 3 \cdot 2^j)$$

$$\sum_{i=1}^3 \sum_{j=1}^2 (i - j)$$

$$\sum_{i=0}^2 \sum_{j=0}^3 i^2 j^3$$

What are the values of the following products?

$$\prod_{i=0}^{10} i$$

$$\prod_{i=5}^8 i$$

$$\prod_{i=1}^{100} (-1)^i$$

$$\prod_{i=0}^{10} 2$$